

MTH2021 Mathematical Methods 2

Self-study sheet 5

1. Find the residues of $f(z)$ at its poles, where

(a)

$$f(z) = \frac{1 + z^2}{z(z - 2)^2}$$

(b)

$$f(z) = \frac{\exp z}{(z - i)^3}$$

(c)

$$f(z) = \frac{1 - \cos z}{z^5}$$

2. Evaluate the integrals

(a)

$$\int_{-\infty}^{\infty} \frac{x \sin x}{(a^2 + x^2)(b^2 + x^2)} dx \quad \text{with } a > 0, b > 0$$

(b)

$$\int_{-\infty}^{\infty} \frac{\cos(ax)}{(1 + x^2)^2} dx \quad \text{with } a > 0$$

(c)

$$\int_{-\infty}^{\infty} \frac{1 + x^2}{1 + x^4} dx$$

3. (a) Use the Darboux inequality to place an upper bound on

$$\left| \int_{C_R} \frac{z^2}{(1 + z^2)^2} dz \right|$$

where C_R is the semicircle of radius R centered at the origin, in the upper half plane. Show that the integral vanishes in the limit $R \rightarrow \infty$.

(b) Evaluate

$$\int_{-\infty}^{\infty} \frac{x^2}{(1 + x^2)^2} dx$$

4. Evaluate the integral

$$\int_0^{2\pi} \frac{d\theta}{5 - 3 \cos \theta}$$
